

RILEY 10 WSW, OR, USA

WMO: 727010

Lat: 43.4711N

Lon: 119.6917W

Elev: 4583

StdP: 12.42

Time Zone: -8.00 (NAP)

Period: 03-19

WBAN: 04128

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-0.9	5.9	-6.4	4.6	-0.3	-0.3	6.4	8.9	15.9	36.0	14.2	34.7	2.1	N/A	0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.6	92.2	57.2	89.4	56.5	86.4	55.6	59.2	84.5	58.0	83.6	56.9	82.2	6.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
51.0	65.8	61.6	48.5	59.8	60.7	46.4	55.2	59.7	27.8	84.4	26.9	84.0	26.2	82.0	65.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	12.6	11.0	DB	-9.6	97.2	5.3	2.7	-13.4	99.1	-16.5	100.7	-19.5	102.3	-23.3	104.3	(4)
(5)			WB	-10.7	62.1	5.1	1.9	-14.4	63.5	-17.4	64.6	-20.2	65.6	-24.0	67.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.3	26.4	30.1	37.1	41.5	50.2	58.3	69.8	67.2	57.7	45.3	33.7	25.4	
	DBStd	16.76	9.05	7.47	6.66	7.12	7.50	8.05	6.40	6.38	8.19	7.74	8.75	9.42	
	HDD50	3541	733	558	402	268	93	20	0	1	25	187	492	762	
	HDD65	7534	1198	978	866	704	461	227	29	51	241	612	940	1227	
	CDD50	1830	0	0	1	14	99	270	614	535	255	40	2	0	
	CDD65	347	0	0	0	0	2	27	178	120	20	0	0	0	
	CDH74	6491	0	0	0	8	119	655	2885	2097	693	34	0	0	
	CDH80	2619	0	0	0	0	20	205	1332	851	206	5	0	0	
Wind	WSAvg	4.9	4.0	4.6	5.4	5.9	5.7	5.5	5.4	5.2	4.7	4.5	4.3	4.1	
Precipitation	PrecAvg	10.6	1.1	0.7	0.9	1.1	1.3	0.7	0.4	0.4	0.4	0.8	1.1	1.2	
	PrecMax	16.8	2.6	2.0	1.9	2.1	3.7	1.9	1.5	1.1	1.3	1.9	2.6	3.0	
	PrecMin	5.2	0.1	0.2	0.2	0.4	0.1	0.0	0.0	0.0	0.0	0.2	0.2	0.3	
	PrecStd	3.1	0.8	0.5	0.5	0.5	0.9	0.5	0.4	0.3	0.4	0.6	0.6	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.4	55.5	66.2	74.8	82.9	91.5	96.4	94.7	89.4	78.4	63.6	49.0	
		MCWB	38.9	42.7	46.3	50.1	54.7	58.8	60.2	57.1	55.5	51.7	47.1	40.6	
	2%	DB	44.7	49.8	60.3	69.4	77.2	85.8	93.1	91.0	85.4	71.9	57.1	45.0	
		MCWB	37.7	40.9	44.0	48.5	51.8	56.7	57.4	56.3	54.2	49.3	44.8	39.6	
	5%	DB	41.7	45.7	55.2	63.6	72.7	81.3	90.7	88.0	82.0	67.0	52.0	41.8	
		MCWB	36.2	38.4	41.9	45.5	50.7	54.8	56.9	55.7	53.5	47.8	42.6	37.2	
	10%	DB	38.9	42.0	50.7	58.2	68.1	77.2	87.9	85.2	77.9	62.1	47.5	38.9	
		MCWB	34.6	36.8	40.8	43.1	48.9	52.8	56.2	55.1	52.1	46.0	40.1	35.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.4	45.0	48.3	51.4	55.9	61.5	61.5	59.4	58.5	53.9	49.4	43.0
			MCDB	46.6	53.0	61.2	71.1	72.5	83.8	89.5	85.3	77.8	69.7	59.7	46.9
2%		WB	39.2	42.4	45.3	49.2	53.5	58.8	59.4	58.1	56.6	51.6	46.3	40.3	
		MCDB	44.0	49.3	55.2	66.7	70.9	82.5	85.5	84.8	79.5	68.1	55.7	43.8	
5%		WB	37.1	40.0	43.3	46.5	51.8	56.6	58.3	57.1	54.9	49.4	44.0	37.7	
		MCDB	41.6	45.7	52.8	61.0	67.3	77.6	84.8	83.9	78.3	62.6	51.8	41.2	
10%		WB	35.3	37.5	41.4	44.1	50.1	54.5	57.1	56.0	53.1	47.2	41.2	35.1	
		MCDB	39.3	42.2	50.0	56.0	64.5	74.2	83.6	82.2	76.2	58.5	47.4	38.6	
Mean Daily Temperature Range	5% DB	MDBR	17.8	18.4	22.1	25.9	26.9	29.3	32.6	32.5	32.3	27.8	21.6	17.7	
		MCDBR	21.2	22.7	31.2	37.1	35.5	35.3	34.4	35.5	38.6	36.1	27.8	18.8	
		MCWBR	14.8	14.0	17.5	18.8	15.6	13.9	12.7	12.8	16.9	19.1	17.4	13.8	
	5% WB	MCDBR	19.5	20.3	27.1	32.3	29.2	32.7	32.3	32.9	35.4	30.5	26.9	16.4	
		MCWBR	14.2	13.5	15.6	17.2	13.3	12.7	12.4	12.2	15.3	16.7	16.6	12.4	
Clear-Sky Solar Irradiance	taub	0.254	0.261	0.270	0.285	0.299	0.302	0.307	0.301	0.285	0.279	0.258	0.253		
	taud	2.593	2.619	2.618	2.564	2.540	2.551	2.545	2.555	2.587	2.601	2.616	2.587		
	Ebn at Noon	289	305	313	313	308	306	304	303	303	292	285	279		
	Edh at Noon	20	23	26	30	32	32	31	30	27	23	19	18		
All-Sky Solar Radiation	RadAvg	532	830	1233	1700	2043	2384	2477	2147	1663	1079	628	427		
	RadStd	80	100	104	145	151	119	90	102	88	85	56	71		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+137	N/A

Nomenclature: See separate page